

System Architecture v3 - Current State

Version:	v3.0
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Status:	CURRENT - USE THIS VERSION

Overview

Current production architecture as of January 2024.

NOTE: John added comments in red on pages 4-6. Sarah incorporated most of them.

Diagram in section 3 still needs updating after the Redis cluster change in Dec.

Components

- + API Gateway: AWS API Gateway + Lambda authoriser
- + Auth Service: Keycloak on EKS (v21)
- + Product Service: FastAPI on EKS (Python 3.11)
- + Order Service: Node.js 20 on EKS
- + Notification Service: NEW - added Dec 2023 (see Sarah's notes)
- + Databases: PostgreSQL 15 (RDS Multi-AZ)
- + Cache: Redis 7 (ElastiCache cluster - 3 nodes)

Changes from v2

- + Added Notification Service (async, SQS-backed)
- + Redis upgraded to cluster mode
- + Moved secrets to AWS Secrets Manager

Diagram

See attached Visio file (Architecture_v3_diagram_FINAL2.vsd).

WARNING: diagram may be out of date - check with John.

